

UNIT 3: CREATIVE INVENTIONS

Chapter 17: Inventions-Related Words & Webquest 5: Robots (Page 136)

This master revision guide is designed for Tunisian Baccalaureate students (4th Year Secondary Education). It comprehensively consolidates vocabulary, advanced grammar rules, phonetics, model argumentative essays, and active recall exercises. It is fully aligned with the official Skills for Life textbook.

Subject: Secondary English (Baccalaureate Level - Tunisia)

Unit: Unit 3 — Creative Inventions

Chapter: Chapter 17 & Webquest 5 (Robotics and Technology)

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PART 1: THE EXHAUSTIVE COURSE THEORY

An academic breakdown of key linguistic milestones required for the Bacculaureate examination.

A. Inventions & Robotics Vocabulary Portfolio

Key Word / Collocation	Part of Speech	Bac Exam Context & Synonyms
breakthrough	Noun	A major scientific or technological discovery. Synonym: landmark discovery.
patent	Noun / Verb	An official license securing sole rights to an invention. Example: to file a patent.
state-of-the-art	Adjective	Incorporating the newest ideas and most advanced technology. Synonym: cutting-edge.
device	Noun	A tool, machine, or piece of equipment designed for a specific task. Synonym: gadget.
revolutionize	Verb	To change something radically or fundamentally. Example: Robots revolutionize medicine.
autonomous	Adjective	Having the freedom or power to act independently without human intervention.
innovative	Adjective	Featuring new, original, and highly creative methods or ideas.

B. Advanced Grammar: Relative Clauses & Pronouns

1. Defining Relative Clauses: Essential to the meaning of the sentence. No commas are used. The pronoun 'that' can replace 'who' or 'which'.

Example: The student who designed the autonomous robot won the prize.

2. Non-Defining Relative Clauses: Extra information that can be omitted. Commas are mandatory. The pronoun 'that' cannot be used as a relative pronoun here.

Example: This state-of-the-art robot, which was patented last month, can perform surgeries.

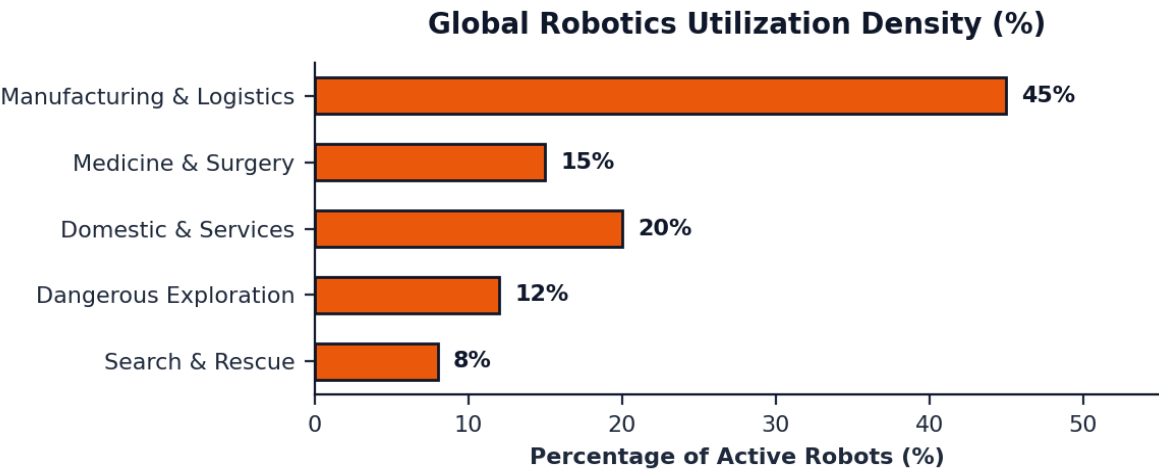
Relative Pronouns Syntactic Grid

Relative Pronoun	Syntactic Function	Bac Exam Rule & Application
Who	Subject (People)	Replaces the person performing the action. (The inventor who made it...)
Whom	Object (People)	Used after prepositions in formal writing. (The engineer to whom we spoke...)
Which	Subject / Object (Things)	Refers to animals or inanimate items. (The patent which was filed...)
Whose	Possessive	Indicates ownership for people, animals, or things. (The robot whose battery died...)
That	Subject / Object	Only used in Defining Relative Clauses for people/things. (The device that works...)

C. Phonetics & Word Stress Rules

1. Pronunciation of past tense '-ed' endings:
- /ɪd/: After /t/ or /d/ sounds (e.g., invented, patented, created).
 - /t/: After voiceless sounds like /p/, /k/, /f/, /s/, /ʃ/, /tʃ/ (e.g., revolutionized, launched).
 - /d/: After voiced sounds like /b/, /g/, /v/, /z/, /m/, /n/ or vowels (e.g., designed, automated, performed).
2. Word Stress Shift: Note the stress shifts during noun derivation:
- IN-vent (Verb - stressed on 1st syllable)
 - in-VEN-tion (Noun - stressed on the syllable before the suffix '-tion')

D. Global Robotics Integration Statistical Analysis



PART 2: THE ACTIVE RECALL REPOSITORY

Review vocabulary, grammar, and phonetics actively. Cover the right column to test your knowledge!

Bac Exam Question Type	Detailed Model Answer & Explanations
Q1: Define a 'breakthrough'.	A major, revolutionary scientific or technological discovery.
Q2: What is the noun for 'to patent'?	The noun is 'patent' (same form) or 'patentation' (process).
Q3: What does 'autonomous' mean?	Able to operate and make decisions independently without human intervention.
Q4: Give a synonym of 'cutting-edge'.	State-of-the-art, ground-breaking, or highly innovative.
Q5: Translate 'brevet' into English.	Patent.
Q6: What is a 'gadget'?	A small mechanical or electronic device, often ingenious or novel.
Q7: Give the verb of 'invention'.	To invent.
Q8: Give the adjective of 'revolution'.	Revolutionary (highly original or completely changing a system).
Q9: What does 'state-of-the-art' mean?	Using the absolute newest and most advanced technologies available.
Q10: Complete: 'To ___ a patent'.	To file a patent / To register a patent / To obtain a patent.
Q11: Translate 'percée scientifique'.	A scientific breakthrough.
Q12: Define 'innovation'.	The practical introduction of a new device, method, or creative idea.
Q13: Give the noun of 'autonomous'.	Autonomy (independence, self-governance).
Q14: Explain the term 'Webquest'.	An inquiry-oriented digital task where learners use internet resources.
Q15: Give a synonym for 'device'.	Apparatus, machine, tool, instrument, or gadget.
Q16: When do we use commas in relative clauses?	In non-defining relative clauses (which provide non-essential extra info).
Q17: Can 'that' be used in non-defining clauses?	No, 'that' is strictly forbidden in non-defining relative clauses.

Q18: Correct: 'The inventor, that won, was happy.'	Correction: 'The inventor, who won, was happy.' (Use who with commas).
Q19: Which pronoun indicates possessive relation?	Whose (e.g., 'The robot whose arm was damaged...').
Q20: Correct: 'The robot whom we bought is fast.'	Correction: 'The robot which/that we bought is fast.' (Whom is for people).
Q21: Combine: 'The man won. He invented it.'	The man who invented it won. (Defining Relative Clause)
Q22: Combine: 'The lab is huge. We work there.'	The lab where we work is huge. (Or: The lab in which we work...)
Q23: What makes 'that' useful in defining clauses?	It can refer to both people and things, replacing who or which.
Q24: Correct: 'The patent whose they filed got approved.'	Correction: 'The patent which they filed got approved.' (Whose is possessive).
Q25: Explain when 'whom' is required.	Whom is used for the object of a verb or preposition (formal style).
Q26: Combine: 'I met Mrs. Heron. Her school is special.'	I met Mrs. Heron, whose school is special.
Q27: Can we omit the relative pronoun?	Yes, in defining clauses, only when the pronoun acts as the object.
Q28: Is this defining? 'This laptop, which is new, is mine.'	No, it has commas, making it non-defining (extra information).
Q29: Correct: 'The car what he designed is electric.'	Correction: 'The car which/that he designed is electric.'
Q30: Complete: 'The scientist ___ patent expired.'	Whose (possessive relation: 'whose patent expired').
Q31: How is the '-ed' pronounced in 'invented'?	/ɪd/ (since the root verb ends with the 't' sound).
Q32: Pronounce the '-ed' in 'designed'.	/d/ (voiced sound 'n' precedes it).
Q33: Pronounce the '-ed' in 'launched'.	/t/ (voiceless sound 'ch' precedes it).
Q34: Where is the stress in 'invention'?	On the second syllable (in-VEN-tion).
Q35: Where is the stress in the verb 'invent'?	On the second syllable (in-VENT).

Q36: Pronounce the '-ed' in 'patented'.	/ɪd/ (root verb ends in 't').
Q37: Pronounce the '-ed' in 'revolutionized'.	/d/ (voiced 'z' sound precedes it).
Q38: Is the '-ed' in 'created' /t/, /d/, or /ɪd/?	/ɪd/ (since create ends in a 't' sound).
Q39: Where is the stress in 'scientific'?	On the third syllable (sci-en-TIF-ic - immediately before '-ic').
Q40: Pronounce the '-ed' in 'developed'.	/t/ (voiceless 'p' sound precedes it).

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PART 3: EXAM EVALUATION QUIZ & REVISION MEMOS

A mini-mock exam evaluating relative clauses, passive forms, and phonetic rules.

1. Choose the correct relative pronoun:
• The robotic machine, _____ was deployed in Tokyo, performs automated operations.
A) who B) which C) that
2. Identify the correct '-ed' pronunciation of 'revolutionized':
A) /ɪd/ B) /t/ C) /d/
3. Complete the sentence with the correct preposition:
• Mrs. Heron aims _____ boosting school attendance through personalized education.
A) on B) at C) in
4. State whether True or False:
• The relative pronoun 'that' can be used directly after a comma in relative clauses. (True / False)
5. Combine the following sentences:
• The patent got approved. The patent protects the autonomous vacuum's technology.

Question No.	Detailed Pedagogical Correction & Rule Proof
1	b) which. Comma indicates a non-defining clause, meaning 'that' cannot be used. 'Which' is the correct pronoun for things.
2	C) /d/. The root verb 'revolutionize' ends in a voiced 'z' sound, making the '-ed' suffix vocalize as /d/.
3	B) at. The verb 'aim' takes the dependent preposition 'at' (aim at + V-ing) to express purpose or objective.
4	False. 'That' is strictly prohibited in non-defining clauses, which are separated by commas.
5	'The patent, which protects the autonomous vacuum's technology, got approved.' (Commas are required because it is a non-defining relative clause).

Revision Memo: Relative Clauses Cheat Sheet

Relative Clause Type	Commas Used?	Pronoun 'That' Allowed?	Main Purpose
Defining	No Commas	Yes, allowed	Provides essential identification details.
Non-Defining	Yes, between commas	No, prohibited	Provides extra non-essential details.